Weighted degree-one doubling

Weighted doubling: local field preservation

z,w,y,2 nonzero + normalized point equation

Normalize literal output → u=(X-xx)², v=sX+(yy-s·xx)

s=f′(xx)/(2yy); explicit Taylor quotient → u divides f-v²

y=0 uses earlier inverse branch. Zero derivative is permitted.

Seven Lean declarations compile. Separate controls and axiom readback return exit zero. The weighted field formula preserves normalized Mumford divisibility on $f = X^5 + a_3X^3 + a_2X^2 + a_1X + a_0$ under the displayed assumptions.

Construction The output has $z' = z$, $w' = 2y$. Its normalized quadratic is $(X - xx)^2$ and its linear polynomial is the tangent interpolant. An explicit quotient identity proves divisibility; input point validity and the tangent equation cancel the residual terms.

Branches and controls Self-add with nonzero ordinate reaches doubling; zero ordinate reaches the earlier inverse branch. A nonunit-weight fixture instantiates the full theorem. A separate zero-derivative fixture also instantiates it. Derivative nonvanishing is not a required denominator condition.

Scope This is field polynomial validity. Global nonsingularity and class-sum interpretation are separate; UInt64 execution, counters, compiler and walk closure are not proved. The curve preprocessing is not included. The guard lemma covers self-add, not every differently weighted equal-point input. The first algebra fixture uses $f = X^5$ and is not presented as a smooth curve.

Evidence Initial failed attempt retained; corrected modules, controls and readback pass. Only standard Lean axioms occur. REPORT.md records exact formula/source boundaries and literature equation(3); sources.json and the manifest bind artifacts. Independent review pending. No scientific trials or repository edits.